MAIC NEXUS CHALLENGE 2026

Technical Architecture

Secure, evidence-aware digital twin architecture for auditable energy analysis and investment decisions.

Team EnerGen AI · Project Irene
T1 — AI for Clean Energy

IRENE SYSTEM ARCHITECTURE

From source files to verifiable energy decisions

01
Ingest
CSV / Excel, documents, images and CAD files

02
Govern
Local parsing + consented recognition, mapping, quality + confirmation

03
Baseline
Confirmed tables + monthly consolidation

04
Simulate
Efficiency, PV and storage design

05
Optimise
Nine-tool local agent + loss-aware actions

06
Verify
Audit package + M&V deterministic fallback

Every output retains source hash, field mapping, unit, evidence class and confirmation status.

Evidence links every asset and value

Entities link physical scope, operating state, engineering constraints and the evidence behind every value.

BUILDING — 6,231 m² · hours → **ZONE** — Use · occupancy → **SYSTEM** — HVAC · electrical → **ASSET** — Power · efficiency → **METER** — State · source

The reference case links source files, field mappings and quality records to the DB evidence graph, four meters and 106.14 kWp of installed PV.

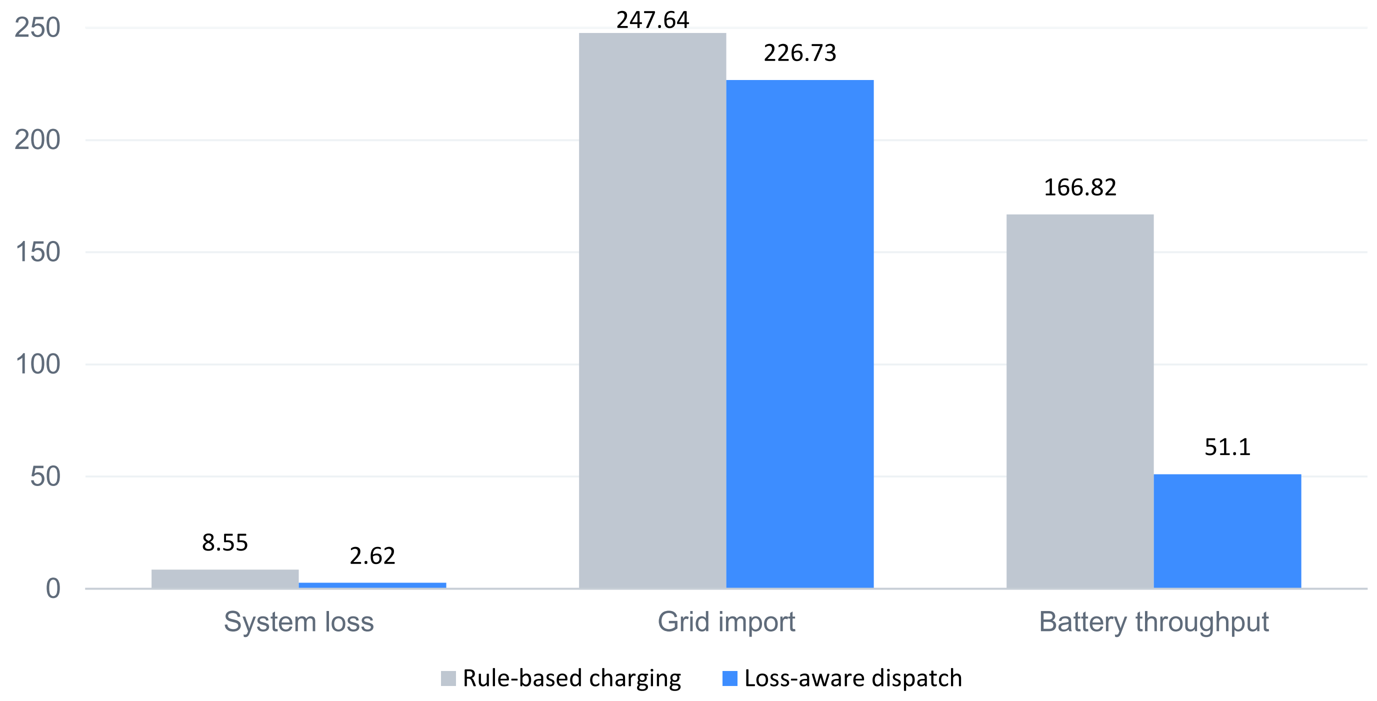
Evidence quality governs every model layer

MODEL LAYER	AVAILABLE EVIDENCE	DATA REQUIREMENT	VALIDATION CHECK
Energy baseline	12 measured monthly anchors	15-min meter + schedules	Month sums + residuals
Forecasting	Calendar prior · 14.36% NMAE	Interval load + indoor sensors	Time split · MAE · RMSE
Digital twin	DB graph + confirmed client project	Zone/circuit/equipment map	Source trace + mapping + constraints
Optimisation	Storage dispatch optimisation	BESS design + supplier quote	Cost · loss · SOC · violations

Evidence boundary: 14.36% NMAE validates the reference time-shape only; DB hourly load remains estimated until interval data are available.

Loss-aware optimisation reduces storage losses

Design study only: DB has no installed battery. Both strategies use identical constraints and terminal state of charge.



69.4%

lower conversion loss

69.4%

less battery throughput

8.4%

less grid import

One governed stack supports audit, pilot and portfolio scale

01

FILE-BASED DIAGNOSIS

Client files → confirmation gate
Auditable baseline + action report
One-off diagnosis fee

FIRST PAID PRODUCT

02

CALIBRATED PILOT

Temporary meters / inverter feeds
Calibration → savings verification
Implementation + M&V fee

PILOT + M&V

03

PORTFOLIO SUBSCRIPTION

Multi-site governed analytics
Portfolio monitoring + benchmarks
Annual subscription

RECURRING VALUE

Target customers: commercial buildings · campuses · industrial parks · energy service companies (ESCOs). Local-first by default; client files stay governed.

Measured first. Optimised second.

Validated on a Ningbo reference case; designed for configurable deployment in Malaysia, pending local pilot validation.

IMPLEMENTATION PATH

File audit → temporary metering → calibration → savings verification → multi-site scale.